Locked tromino tilings of rectangles

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Abstract

A tromino is a set of three grid cells connected through shared sides. A tiling of a grid by trominoes is locked if no two of its tiles can be removed and their six cells covered again by two trominoes in a different way, and the grid has at least one other tromino tiling. A locked tiling is a state that no step of the recombination (ReCom) Markov chain used in redistricting can change, on a grid with districts of three cells. Tucker-Foltz (2023) showed that no $m \times n$ grid with $\min(m, n) \leq 3$ has a locked tromino tiling, constructed locked tilings whenever $3 \mid m$, $m \geq 6$ and $n \geq 100$, and conjectured that $n \geq 4$ suffices. We prove this conjecture. Together with his theorem it follows that an $m \times n$ rectangle has a locked tromino tiling if and only if $3 \mid mn$ and $\min(m, n) \geq 4$. As in his proof, we glue explicit locked blocks, mostly of heights 6 and 9, along full rows and columns; the new ingredients are a pair of blocks for each width from 4 to 55 (with heights 9 and 12, and two single tilings, at width 5) and four blocks of widths 8 and 9 for all larger widths. The gluing argument and every block are checked in Lean 4.

The proofs, the Lean formalization, the certificate checkers and the text were produced by AI models (from OpenAI and Anthropic), working under the author's direction.

Status: Checked in Lean from the definitions (Theorem 2); not yet reviewed by experts. Theorem 1 also uses Theorem 3.1 of Tucker-Foltz's preprint, by a short written argument.

1 Introduction

Cut a rectangular grid of cells into *trominoes*, sets of three cells connected through shared sides: straight ones and L-shaped ones, in any orientation. Pick two of the trominoes, remove them, and try to cover the six freed cells with two trominoes in a different way. If this is possible the change is a *recombination move* (Figure 2). A tiling is *locked* if no recombination move is possible, for any two of its tiles, although the grid does have other tromino tilings (Figure 1). We determine exactly which rectangles have a locked tromino tiling.

Theorem 1. *Let m and n be positive integers. The $m \times n$ rectangle has a locked tromino tiling if and only if 3 divides mn and $\min(m, n) \geq 4$.*

Tucker-Foltz introduced locked tilings [7] as obstructions for the ReCom Markov chain of DeFord, Duchin and Solomon [4], which samples redistricting plans by repeatedly merging two districts and splitting the union into two new ones. When the map is an $m \times n$ grid graph whose vertices have equal population and every district has three vertices, the plans are the tromino tilings, a ReCom step is a recombination move, and a locked tiling is a plan that no ReCom step can change. Tucker-Foltz asked for which grids and which tile sizes locked tilings exist. For trominoes he proved that none exist when one side of the rectangle is at most 3 [7, Theorem 3.1], which with the area

count is the "only if" direction of Theorem 1. He found that the 6×4 rectangle has 7 locked tilings up to symmetry and the 6×6 rectangle has 50 locked tilings, assembled copies of a 6×6 locked tiling into locked tilings of every rectangle whose sides are multiples of 6, proved that locked tilings exist whenever $3 \mid m$, $m \geq 6$ and $n \geq 100$ [7, Theorem 3.2], and conjectured that $n \geq 4$ suffices [7, Section 3]. We prove his conjecture, which is the "if" direction.

Theorem 2 (conjectured by Tucker-Foltz)**.** *Let m and n be integers with $3 \mid m$, $m \geq 6$ and $n \geq 4$. Then the $m \times n$ rectangle has a locked tromino tiling.*

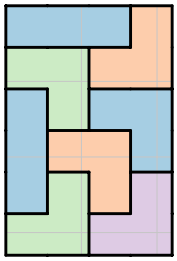


Figure 1: A locked tromino tiling of the 6×4 rectangle. Thick lines are tile boundaries; colors only separate neighboring tiles. It is the height-6 block of width 4 used in the proof.

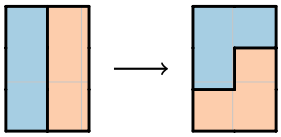


Figure 2: A recombination move: two straight trominoes side by side cover the same six cells as two L-trominoes. A locked tiling has no pair of tiles that allows such a move.

Up to transposition, the smallest rectangles allowed by Theorem 1 are 6×4, 6×5, 6×6 and 9×4; among them, only the 6×4 and 6×6 cases were settled before [7]. Theorem 1 is a statement about this grid model with districts of three cells: it shows exactly where such isolated plans exist, and says nothing about districts of realistic size or about redistricting software in practice. Related work establishes irreducibility of ReCom, that every plan can reach every other, in other regimes: on regions of the triangular lattice and of the square grid when district sizes may differ slightly from equal (Cannon [3]; Akitaya, Dituro, Gonczi, Korman, Souvaine, Stock and Tóth [1]), and in some exactly balanced cases, among them a $3 \times n$ grid cut into three districts (Emre, Jerison and Veomett [5]).

Aanjaneya and Pal [2] earlier proved that every rectangle with $m, n \geq 4$ and $3 \mid mn$ admits a fault-free tiling by L-trominoes [2, Theorem 2]: the same rectangles as in Theorem 1, for a different property. Like ours, their construction builds large rectangles from finitely many small ones by adding rows and columns. A fault line is a full horizontal or vertical cut that crosses no tile; fault-free means no such cut exists. This is a different condition from being locked, which forbids every two-tile recombination and permits both straight and L-trominoes. In particular, the full cuts between our glued blocks are fault lines, although the resulting tilings are locked. Their fault-free existence theorem therefore does not establish the locked-tiling claim proved here.

What is new, and the hardest step. Tucker-Foltz's Theorem 3.2 already proves Theorem 2 for $n \geq 100$, by the method we use: cut the grid along full rows and columns into blocks of heights

6 and 9, and fill the blocks with locked tilings that stay locked across every shared side. His blocks are assembled from strips of lengths 22, 19 and 8, which gives the bound $n \geq 100$. What we add are all the remaining cases, those not covered by his Theorem 3.2 in either orientation or by his 6×6 copies; each has a side shorter than 100. This makes the theorem sharp. For each width from 4 to 55 we give two blocks, of heights 6 and 9, that can be stacked in either order and on copies of themselves; at width 5 the heights are 9 and 12, and the 6×5 and 15×5 rectangles, whose heights are not sums of nines and twelves, get single tilings. Widths 56 and up use four blocks of widths 8 and 9, since every integer from 56 on is a sum of eights and nines.

The hardest step is the narrow widths. Tucker-Foltz notes that tromino tilings "are largely unstructured and difficult to find, especially when m or n is odd" [7, Section 3], and a block must be locked inside and stay locked against every block it can meet, in both orders and against a copy of itself. The gluing lemma itself (Lemma 7) is elementary: when blocks meet along full rows and columns, only tiles in one block or across one shared side can recombine. Its use is that it turns each width into a finite problem, two blocks and four seams, which a computer search solves; the search is not part of the proof. The other demanding part is the checking: Lean's kernel checks every block and seam, which needed proved shortcuts to run in minutes (Section 7).

How it is checked. Theorem 2 is proved in Lean 4 [6], from definitions of trominoes, tilings and recombination that are written out in a few lines (Section 7). The proof includes the gluing argument, the arithmetic, and the check of every block and every shared side, carried out by Lean's kernel. Two separately written programs check the same blocks again. The deduction of Theorem 1 in Section 3 is a written proof that uses Tucker-Foltz's Theorem 3.1, which is not formalized.

Map of the paper. Section 2 gives the definitions and two basic facts. Section 3 derives Theorem 1 from Theorem 2 and Tucker-Foltz's Theorem 3.1. Section 4 outlines the proof of Theorem 2, Section 5 proves the gluing lemma and Section 6 presents the blocks and completes the proof. Section 7 states what Lean and the programs check, and Appendix A lists the blocks.

2 Definitions and two basic facts

This section fixes the definitions, shows they agree with Tucker-Foltz's, and proves two facts used throughout: tiles that do not touch can never recombine, and the rectangles of Theorem 2 have more than one tiling.

A *cell* is a pair of integers (i, j), read as row i and column j. Two cells are *adjacent* if they agree in one coordinate and differ by one in the other. The $m \times n$ *rectangle* is the set of cells (i, j) with $0 \leq i < m$ and $0 \leq j < n$; it has m rows and n columns. A *tromino* is a set of three cells that is connected: any two of its cells are joined by a sequence of adjacent cells inside it. Up to translation there are six trominoes, two straight and four L-shaped. A *tiling* of a set of cells is a partition of it into trominoes, called its *tiles*.

Definition 3. Let T be a tiling and A, B two of its tiles. A *recombination move* on $\{A, B\}$ replaces them by two disjoint trominoes C, D with $C \cup D = A \cup B$ and $\{C, D\} \neq \{A, B\}$. The pair $\{A, B\}$ is *rigid* if it admits no recombination move. The tiling T of a rectangle is *locked* if every pair of its tiles is rigid and the rectangle has a tiling other than T.

Tucker-Foltz [7, Section 1] works with the *metagraph* $M(G, 3)$ of a graph G: its vertices are the partitions of G into connected subgraphs of three vertices each, and two partitions are adjacent when they differ only on two parts. A locked tiling is an isolated vertex of $M(G, 3)$ that is not its

only vertex. For the grid graph G of a rectangle, the vertices are the tromino tilings, and two tilings that differ only on two parts cover the same six cells with those parts, so its edges are exactly the recombination moves. The two definitions of a locked tiling therefore agree.

Translating a tiling, or reflecting it in a horizontal, vertical or diagonal line, preserves adjacency, so it maps trominoes to trominoes, recombination moves to recombination moves and locked tilings to locked tilings. In particular the transpose $(i, j) \mapsto (j, i)$ turns locked tilings of the $m \times n$ rectangle into locked tilings of the $n \times m$ rectangle.

Lemma 4. *Let A and B be disjoint trominoes such that no cell of A is adjacent to a cell of B. Then $\{A, B\}$ is rigid.*

Proof. Let C and D be disjoint trominoes with $C \cup D = A \cup B$. A path of adjacent cells inside $A \cup B$ that starts in A never reaches B, because its first step into B would join a cell of A to an adjacent cell of B. Since C is connected, it therefore lies inside A or inside B, and since all three sets have three cells, C equals A or B. The same holds for D, and $C \neq D$ because they are disjoint and nonempty. Hence $\{C, D\} = \{A, B\}$. □

Lemma 5. *If m and n are positive integers with $3 \mid m$ and $n \geq 3$, the $m \times n$ rectangle has at least two tromino tilings. Hence a tiling of it is locked as soon as every pair of its tiles is rigid.*

Proof. Cover every column by vertical straight trominoes; or cover the first three columns by horizontal straight trominoes, one in each row, and the remaining columns by vertical ones (Figure 3). The second tiling contains the tile $\{(0, 0), (0, 1), (0, 2)\}$ and the first does not. Any tiling differs from at least one of the two. □

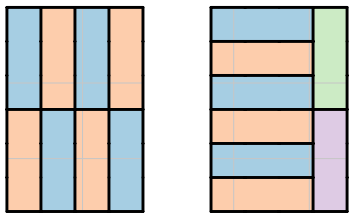


Figure 3: Two tilings of the 6×4 rectangle (Lemma 5). Every rectangle with $3 \mid m$ and $n \geq 3$ has these two, so a tiling all of whose pairs are rigid is locked.

3 The classification from the conjecture

This section proves Theorem 1, assuming Theorem 2. The "only if" half rests on Tucker-Foltz's theorem for narrow grids, quoted here in his notation, where $G(m, n)$ is the $m \times n$ grid graph.

Theorem 6 (Tucker-Foltz [7, Theorem 3.1])**.** *For any positive integers m and n such that $m \leq 3$, $M(G(m, n), 3)$ is connected.*

Proof of Theorem 1. Suppose first that the $m \times n$ rectangle has a locked tiling T. Its mn cells are split into tiles of three cells, so $3 \mid mn$. Suppose that $\min(m, n) \leq 3$. Transposing if necessary, we may assume $m \leq 3$; then $M(G(m, n), 3)$ is connected by Theorem 6. But T is an isolated vertex of this graph, and the graph has another vertex, so it is not connected. This contradiction shows $\min(m, n) \geq 4$.

Conversely, suppose $3 \mid mn$ and $\min(m, n) \geq 4$. Since 3 is prime, it divides m or n; transposing if necessary, $3 \mid m$. As $m \geq 4$ and m is a multiple of 3, $m \geq 6$. Also $n \geq 4$, so Theorem 2 gives a locked tiling of the $m \times n$ rectangle, and transposing back gives one of the original rectangle. □

4 Overview of the proof of Theorem 2

By Lemma 5 it is enough to tile the $m \times n$ rectangle so that every pair of tiles is rigid. A direct search can do this for one rectangle at a time, but not for infinitely many, so we build the tiling from a few fixed rectangles, the *blocks*, each carrying a tiling all of whose pairs are rigid. Write m as a sum of heights and n as a sum of widths, cut the rectangle along the corresponding full rows and columns, and place a block in each piece. The obvious danger is a pair of tiles from two different blocks that recombines.

The gluing lemma (Lemma 7) shows that this can happen only across a shared side: when the cuts run through the whole rectangle, two tiles in blocks that touch only at a corner, or not at all, have no adjacent cells, and Lemma 4 applies. So it suffices to check each block and each *seam*, the shared side of two blocks placed next to each other, for every pair of blocks that can occur next to each other in either order, including a block next to a copy of itself.

Two systems of blocks then cover all rectangles. Every multiple of 3 from 6 on is a sum of sixes and nines, so blocks of heights 6 and 9 suffice everywhere except at width 5. For each width n from 4 to 55 a single column of blocks of width n suffices; width 5 uses heights 9 and 12 instead, plus single tilings of the 6×5 and 15×5 rectangles, the two heights that are not sums of nines and twelves. Every width from 56 on is a sum of eights and nines, and four blocks, of heights 6 and 9 and widths 8 and 9, fill all those rectangles. Tucker-Foltz's proof of Theorem 3.2 [7] has the same shape, with blocks of heights 6 and 9 assembled from strips of lengths 22, 19 and 8, which reach every width from 100 on.

5 The gluing lemma

This section proves that a tiling assembled from blocks along full rows and columns has only rigid pairs as soon as each block and each seam does.

Let $m = h_1 + \cdots + h_r$ and $n = w_1 + \cdots + w_s$ with positive integers h_k and w_l. Put $H_k = h_1 + \cdots + h_k$ and $W_l = w_1 + \cdots + w_l$, with $H_0 = W_0 = 0$. The *block* (k, l) is the set of cells (i, j) with $H_{k-1} \le i < H_k$ and $W_{l-1} \le j < W_l$, an $h_k \times w_l$ rectangle. The blocks partition the $m \times n$ rectangle. Two blocks are *side neighbors* if they lie in the same row of blocks and in consecutive columns of blocks, or in the same column of blocks and in consecutive rows of blocks.

Lemma 7 (gluing)**.** *For each block (k, l) let T_{kl} be a tiling of it, and let T be the union of these tilings, a tiling of the $m \times n$ rectangle. Suppose that*

(a) every pair of tiles of each T_{kl} is rigid, and

(b) for every two side neighbors (k, l) and (k', l'), every pair consisting of a tile of T_{kl} and a tile of $T_{k'l'}$ is rigid.

Then every pair of tiles of T is rigid.

Proof. Let A and B be tiles of T, in blocks (k, l) and (k', l'). If the blocks are equal or side neighbors, $\{A, B\}$ is rigid by (a) or (b). Otherwise we show that no cell of A is adjacent to a cell of B, so that Lemma 4 applies. Let $(i, j) \in A$ and $(i', j') \in B$ be adjacent. Then $|i - i'| \le 1$, and since every height h_k is positive, the rows i and i' lie in the same row of blocks or in consecutive ones; the same holds for the columns j and j'. Adjacent cells also agree in one coordinate, so the two blocks share a row of blocks or a column of blocks. Hence they are equal or side neighbors, contrary to assumption. □

Condition (b) concerns only two blocks at a time, and rigidity is unchanged by translation, so it can be checked on a fixed pair of rectangles. For a tiling X of an $h \times w$ rectangle and a tiling Y of an $h' \times w$ rectangle, say that X *stacks on* Y if, after Y is moved down by h rows, every tile of X forms a rigid pair with every tile of the moved Y; the line between them is the seam. Similarly, for equal heights, X *fits left of* Y if this holds after Y is moved right by the width of X. A block is *rigid* if all pairs of its tiles are. Figure 4 shows the four stacks of the two width-4 blocks, and Figure 5 the four blocks for large widths, with both kinds of seam.

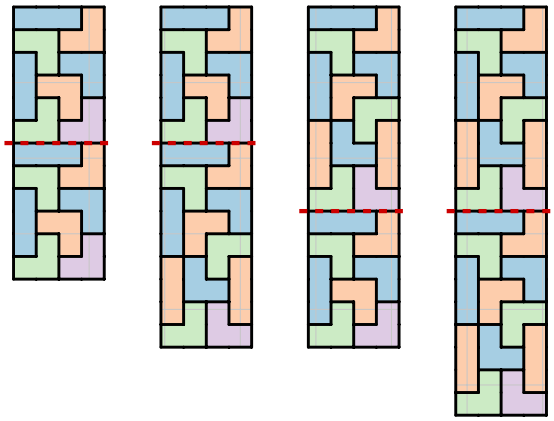


Figure 4: The two blocks of width 4, of heights 6 and 9, stacked in all four orders: 6 on 6, 6 on 9, 9 on 6 and 9 on 9. Each dashed red line is a seam. No two tiles recombine, inside a block or across its seam, so any column of these blocks is a locked tiling.

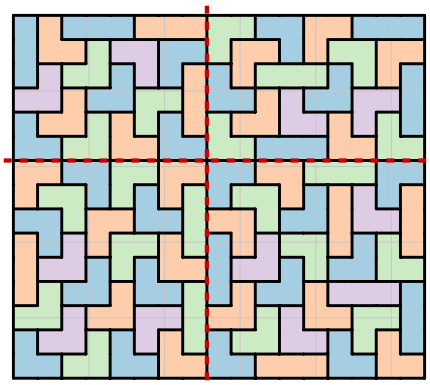


Figure 5: The four blocks for widths $n \geq 56$, of heights 6 and 9 and widths 8 and 9, here in a 15×17 rectangle with the horizontal and vertical seams dashed. Each block also stacks on and fits beside every block of the same width or height, including itself, so these four fill every rectangle with $3 \mid m$, $m \geq 6$ and $n \geq 56$.

Corollary 8. *(i) Let S be a set of rigid blocks of a common width w such that every $X \in S$ stacks on every $Y \in S$, including $Y = X$. Then any column of blocks from S, placed one under another, is a tiling of an $h \times w$ rectangle all of whose pairs of tiles are rigid, where h is the sum of the heights.*

(ii) Let $P_{h,w}$, for h in a set $\mathcal{H}$ and w in a set $\mathcal{W}$, be rigid blocks of size $h \times w$ such that $P_{h,w}$ stacks on $P_{h',w}$ and fits left of $P_{h,w'}$ for all $h, h' \in \mathcal{H}$ and $w, w' \in \mathcal{W}$. If $m = h_1 + \cdots + h_r$ with every $h_k \in \mathcal{H}$ and $n = w_1 + \cdots + w_s$ with every $w_l \in \mathcal{W}$, the tiling that places a translate of P_{h_k,w_l} in block (k, l) has only rigid pairs.

Proof. Both are Lemma 7. In (ii), condition (a) is the rigidity of the blocks, and for side neighbors (k, l) above $(k+1, l)$ or left of $(k, l+1)$, condition (b) is the statement that P_{h_k,w_l} stacks on P_{h_{k+1},w_l} or fits left of $P_{h_k,w_{l+1}}$, translated into place. Part (i) is the case of a single column of blocks. □

6 The blocks and the proof of Theorem 2

This section states the finite facts the proof needs, the arithmetic that matches them to all rectangles, and assembles the proof.

Proposition 9. *The following tilings exist, and are given explicitly in the folder* `data/` *(Appendix A).*

(i) For each width w with $4 \le w \le 55$ and $w \ne 5$, rigid blocks $B_{6,w}$ and $B_{9,w}$ of sizes $6 \times w$ and $9 \times w$, each of which stacks on each, including itself.

(ii) Rigid blocks $B_{9,5}$ and $B_{12,5}$ of sizes 9×5 and 12×5, each of which stacks on each, including itself; and rigid tilings of the 6×5 and 15×5 rectangles.

(iii) Rigid blocks $P_{h,w}$ of size $h \times w$ for $h \in \{6, 9\}$ and $w \in \{8, 9\}$, such that $P_{h,w}$ stacks on $P_{h',w}$ and fits left of $P_{h,w'}$ for all $h, h' \in \{6, 9\}$ and $w, w' \in \{8, 9\}$.

Figure 4 shows the blocks of (i) for $w = 4$, Figure 6 the tilings of (ii), and Figure 5 those of (iii).

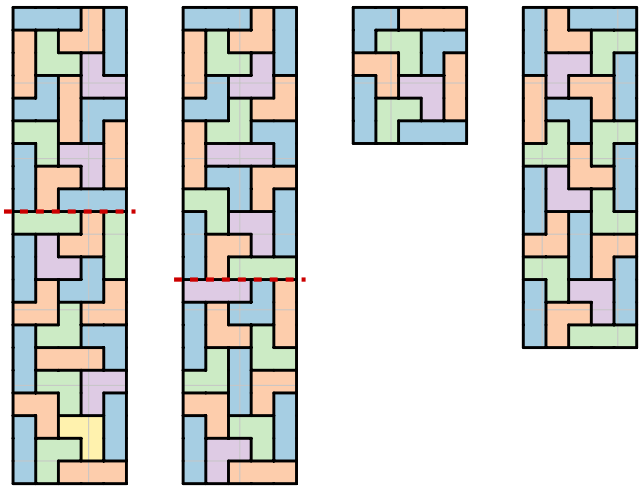


Figure 6: Width 5. Left: the blocks $B_{9,5}$ and $B_{12,5}$ stacked in both orders, seams dashed (each also stacks on itself). Right: the single tilings of the 6×5 and 15×5 rectangles, the two heights that are not sums of nines and twelves.

Proof. Each statement is a finite check of explicit tilings: that each block is a tiling of its rectangle by trominoes, and that each of the listed pairs of tiles, inside a block or across a seam, admits only

its own partition into two trominoes. Lean's kernel checks all of them, and so do two separately written programs (Section 7). □

Lemma 10. *(a) Every multiple of 3 that is at least 6 is $6a + 9b$ for some integers $a, b \geq 0$.*

(b) Every integer that is at least 56 is $8a + 9b$ for some integers $a, b \geq 0$.

(c) Every multiple of 3 that is at least 9, other than 15, is $9a + 12b$ for some integers $a, b \geq 0$.

Proof. (a) Write $m = 3q$ with $q \geq 2$. If q is even, $m = 6 \cdot (q/2)$; if q is odd, then $q \geq 3$ and $m = 9 + 6 \cdot (q-3)/2$. (b) Write $n = 8Q + r$ with $0 \leq r \leq 7$. Then $Q \geq 7 \geq r$ and $n = 8(Q-r) + 9r$. (c) Dividing by 3, we need every integer $q \geq 3$ other than 5 to be $3a+4b$: this holds for $q = 3, 4, 6, 7, 8$, and adding threes reaches every larger integer. □

Proof of Theorem 2. Let $3 \mid m$, $m \geq 6$ and $n \geq 4$. By Lemma 5 it suffices to find a tiling of the $m \times n$ rectangle all of whose pairs of tiles are rigid.

If $n \geq 56$, Lemma 10(a),(b) writes m as a sum of sixes and nines and n as a sum of eights and nines, and Corollary 8(ii) with the blocks of Proposition 9(iii) gives the tiling.

If $4 \leq n \leq 55$ and $n \neq 5$, Lemma 10(a) writes m as a sum of sixes and nines, and Corollary 8(i) with the blocks $B_{6,n}$ and $B_{9,n}$ of Proposition 9(i) gives the tiling.

If $n = 5$ and $m \in \{6, 15\}$, Proposition 9(ii) gives the tiling directly. Otherwise $m \geq 9$ and $m \neq 15$, Lemma 10(c) writes m as a sum of nines and twelves, and Corollary 8(i) with $B_{9,5}$ and $B_{12,5}$ gives the tiling. □

7 Computation and formalization

This section states what is checked, by what, and what is trusted. The files named here are in the folder that accompanies this paper.

The Lean statement. The definitions in `lean/Core.lean` are those of Section 2: a cell is a pair of integers, adjacency is distance one in one coordinate, a tile is a set of three distinct cells joined by walks of adjacent cells inside it, a tiling is a family of tiles that partitions the rectangle, a pair of tiles is rigid if every partition of its union into two tiles is the original pair, and a tiling is locked if all its pairs are rigid and the rectangle has another tiling. The main theorem, `LockedTromino.Witnesses.locked_tromino_rectangles`, states

$$\forall\, m\ n, \quad 6 \leq m \ \to\ 3 \mid m \ \to\ 4 \leq n \ \to\ \exists\, T,\ \mathrm{Locked}\,(\mathrm{Rectangle}\ m\ n)\ T,$$

with no other hypotheses. Lean checks that it depends only on the three axioms of Lean's standard logic: propositional extensionality, the axiom of choice and the soundness of quotients (`propext`, `Classical.choice` and `Quot.sound`). The formal proof follows Sections 2, 5 and 6: Lemma 4, the gluing lemma for arbitrary row and column cuts, Corollary 8, Lemma 10, the two tilings of Lemma 5, and Proposition 9.

The finite checks in Lean. The blocks of Proposition 9 appear in `lean/Generated/` as explicit lists of cells, one file per width. Each finite fact is proved by evaluation in Lean's kernel (`decide`), with no compiled code trusted. To keep this fast, the formal proof establishes a few checking procedures once and for all and then evaluates them: a table assigning each cell its tile proves that the tiles partition the rectangle; listing, for each tile, the tiles that contain a neighbor of one of its

cells reduces rigidity to those pairs, the others being rigid by Lemma 4; and the pairs that touch fall into 162 shapes up to translation and order, each checked once and reused by a proved translation invariance. A wrong cell, owner, neighbor list or shape fails the build.

The programs. Two programs, `checks/locked.py` and `checks/independent.py`, written separately and sharing no code, check every block and seam of Proposition 9 directly from the definitions, by listing all partitions of each six-cell union into two trominoes. A third program, `checks/check.py`, runs both, checks that `data/` holds exactly the blocks and seams the proof needs, checks that the Lean files in `lean/Generated/` are exactly the rendering of `data/`, and checks that both programs reject planted errors. As a control it enumerates all locked tilings of the 6×4 and 6×6 rectangles. It finds 26 locked tilings of the 6×4 rectangle, a count obtained here; their 7 symmetry classes and the 50 tilings of the 6×6 rectangle recover the counts reported by Tucker-Foltz. The figures in this paper are drawn from `data/` by `checks/figures.py`, which also checks that each drawn rectangle, taken as a whole, is a tiling with only rigid pairs.

What is trusted. For Theorem 2: the Lean kernel and the definitions in `lean/Core.lean`. The search that found the blocks is not part of the proof. For Theorem 1, in addition: Tucker-Foltz's Theorem 3.1 and the short written argument of Section 3.

Acknowledgments

Jamie Tucker-Foltz posed the problem, gave the definitions used here, proved the case of narrow grids and the case $n \geq 100$ by the block method we follow, and computed the small cases we use as controls. Lean and its contributors supplied the formal-proof system.

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A The blocks

This appendix describes where the tilings of Proposition 9 are: their coordinates are in the folder `data/` released with this paper, one JSON file per family, and Figures 1, 4, 6 and 5 draw ten of them. Each file names its blocks, gives each block's height, width and tiles (each tile a list of three cells (i, j) with i the row), and lists the seams to be checked: `vertical` $[X, Y]$ means X stacks on Y, and `horizontal` $[X, Y]$ means X fits left of Y.

File in `data/`	Contents	Blocks	Stacks	Side by side
`witnesses/width-w.json` ($w \neq 5$)	$B_{6,w}$, $B_{9,w}$, $4 \leq w \leq 55$	2 each	4 each	–
`witnesses/width-5.json`	$B_{9,5}$, $B_{12,5}$	2	4	–
`witnesses/direct-6x5.json`	a rigid 6×5 tiling	1	–	–
`witnesses/direct-15x5.json`	a rigid 15×5 tiling	1	–	–
`palette.json`	$P_{h,w}$, $h \in \{6, 9\}$, $w \in \{8, 9\}$	4	8	8
Total, 55 files		110	216	8

In all, the proof uses 110 blocks and 224 ordered seams. The seams of each family include both orders of two different blocks and each block against a copy of itself.